Set 22 Statistics Inference

A **point estimate** of a parameter is a single number that serves as an estimate for the true value of the parameter. The estimate comes from a statistic, and is then calculated from the sample data.

An **interval estimate** of a parameter is an interval of real numbers, where each number in the interval is a single number that serves as an estimate for the true value of the parameter. The estimate comes from a statistic, and is then calculated from the sample data.

A **pivotal quantity** is a function of observations with a distribution that does not depend on the value of any unknown parameters.

Example 1: Suppose we have a random sample of size n, either from some normal distribution, or where n is large. Then the following is a pivotal quantity, as it always has standard normal distribution, regardless of what the values of μ, σ are.

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

We can use these pivotal quantities to construct a **random interval** for one of the parameters. A random interval is an interval of real numbers whose endpoints are random variables.

Example 2: Suppose we have a random sample of n observations from some normal population. We can find that

$$P\left(-1.96 \leq \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95$$

This can be rewritten as:

$$P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

We have constructed the random interval

$$\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}}\ ,\ \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

and this interval has the property that over many repetitions of the experiment, 95% of the time the true value of μ will lie within the interval.

Suppose we know the value of σ and carry out the experiment and compute the value of $\overline{x}$. We would call the following *fixed* interval a **95% confidence interval** for population mean μ.

$$\left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}}\ ,\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

The lower end of this interval is the **lower confidence limit** and the upper end is the **upper confidence limit**.

If we were looking to create an interval with a different level of confidence, we could follow the same procedure. We would find the following would be a $100(1-\alpha)\%$ confidence interval for population mean μ:

$$\overline{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

In several different scenarios, the pivotal quantity from which we derive our confidence interval has standard normal distribution (or another symmetric distribution). In all such cases, the confidence interval is of the form:

$$\text{estimate} \pm (\text{critical value})(\text{standard error})$$

Common critical values:

% Confidence	Critical Value
90	$z_{\alpha/2} = z_{0.05} = 1.645$
95	$z_{\alpha/2} = z_{0.025} = 1.96$
99	$z_{\alpha/2} = z_{0.005} = 2.575$

Example 3: Suppose that for 25 people, the average height $\overline{x} = 1.7$ m. Suppose it is known that the standard deviation of people's heights is 0.08 m, and that people's heights are normally distributed. Find a 95% confidence interval for μ, the population's mean height.

Interpretation: We should **not** interpret this as meaning that there is a 95% chance that the true mean height is between 1.66864 and 1.73136 meters.

Instead, we should interpret the idea of "95% confidence" as meaning that if we were to repeat this experiment many times, about 95% of the intervals we would construct would contain the true value of μ.

The level of confidence we have is measuring our confidence in our **procedure**, not in the particular interval we've created.

We should note that the d, the distance from the center to the lower or upper confidence limit (also called the **margin of error**) depends on the critical value and the standard error. The width is $2d$.

$$d = (\text{critical value})(\text{standard error})$$

$$d = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

The critical value increases as the confidence level increases. This means that if we wish to increase our confidence level, the interval will get larger as a result.

Also, note that as the sample size increases, the standard error gets smaller. As a result, by increasing the sample size we can decrease the size of our confidence interval.

If the standard deviation is known (or assumed), and the critical value is fixed in advance, we can determine the sample size needed to give us a particular margin of error, d.

Example 4: Suppose the lifetime of certain type of lightbulb is normally distributed and has a standard deviation of $\sigma = 200$ hours. How many samples do we need to be create a 95% confidence interval for μ, the mean lifespan, with a margin of error of 10 hours (or within 10 hours)?

Example 5: How many samples do we need to estimate the mean with 95% confidence interval for μ with a width of 40 hours?

In most real-life cases, the true values of σ and μ are both unknown. However, if our sample size is quite large, then s, the sample standard deviation, will be approximately equal to σ.

For a large-sample size scenario ($n \geq 40$), we use pivotal quantity

$$Z = \frac{\overline{X} - \mu}{s/\sqrt{n}} \approx N(0,1),$$

and the following is an approximate $100(1-\alpha)\%$ confidence interval for population mean μ:

$$\left(\overline{x} - z_{\alpha/2}\frac{s}{\sqrt{n}}, \overline{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}\right)$$

We call $s/\sqrt{n}$ the **estimated standard error** of the sample mean, since it is an estimate of $\sigma/\sqrt{n}$, the standard error of the sample mean.

Example 6: At a particular location, fifty daily measurements of wind speed (in m/s) are made. It is found that $\overline{x} = 15.9\ m/s$ and $s = 7.7\ m/s$.

Find a 98% confidence interval for μ, the average daily wind speed. Assume that the measurements constitute a random sample from the population of all wind speed measurements.

Example 7: Suppose $X_1, X_2, \ldots, X_n$ are a random sample from a normal population. Also, suppose that the sample size is not large.

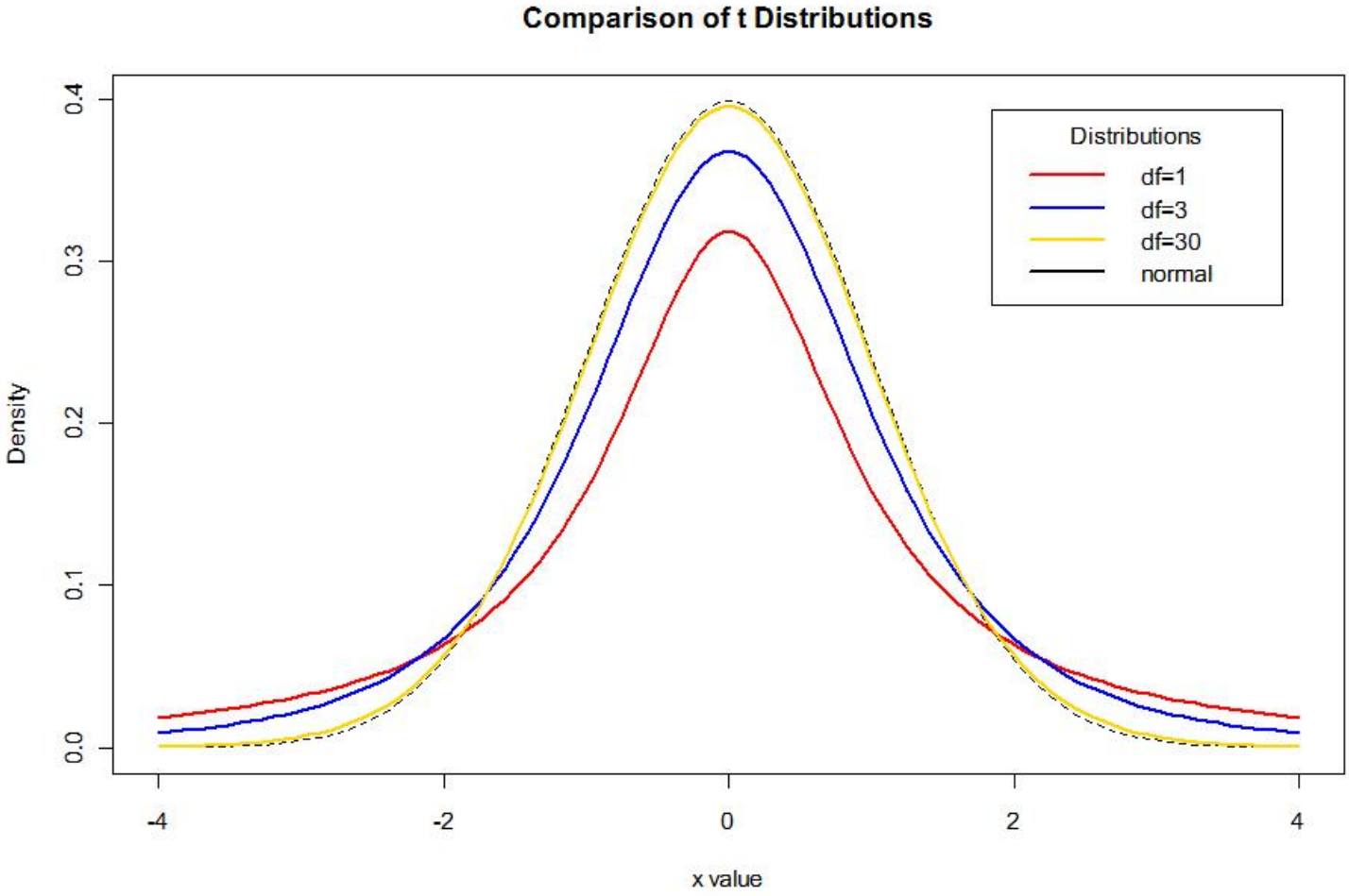
The following pivotal quantity will have t-distribution, with $n-1$ degrees of freedom.

$$t = \frac{\overline{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

Properties of the t distribution:

1. The t distribution is continuous, and defined on $(-\infty, \infty)$.
2. The t distribution is symmetric, bell-shaped, and centered at zero.
3. The number of degrees of freedom affect the shape of the distribution; as the number of degrees of freedom increases, the distribution becomes more peaked, and the tails become thinner.
4. When the number of degrees of freedom is large (30 or more), the t-distribution is approximately a standard normal distribution.

Notation: To denote a t-distribution with k degrees of freedom, we write t_k.


Comparison of t Distributions
Distributions
df=1
df=3
df=30
normal
Density
0.0
0.1
0.2
0.3
0.4
-4
-2
0
2
4
x value

Confidence Interval for the Population Mean: The $100(1-\alpha)\%$ confidence interval for μ, when a **small** sample of size n is taken from a normal distribution (σ is unknown) is:

$$\overline{x} \pm t_{n-1,\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Here, $t_{n-1,\alpha/2}$ is the critical value of the t-distribution with $n-1$ degrees of freedom.

Note: Provided that the population is reasonably close to being normally distributed (i.e. approximately normal, or near-normal), the above can still be used.

We only need to use the above for small sample sizes. (If the sample size was large, then critical value from t_{n-1} and the critical value from Z would be nearly identical.)

Example 8: The following data is collected on the mass (in grams) of adult white mice.

$$14.6, 13.2, 19.5, 10.1, 8.8, 15.5, 16.1$$

Assuming that the weights of mice are normally distributed, find a 95% confidence interval for μ, the mean weight of adult white mice.

Example 9: A random sample of nine observations on the number of pounds of nitrogen created per duck farm per day is taken:

$$4.9, 5.8, 5.9, 6.5, 5.5, 5.0, 5.6, 6.0, 5.7$$

Assuming that the number of pounds is approximately normally distributed, find a 99% confidence interval for μ.

Suppose we make n observations from some binomial experiment and n is large. The following has approximately standard normal distribution, regardless of the value of the parameter p. Recall that $\widehat{p}$ is the sample proportion, which we use as our **estimate** for p.

$$Z = \frac{\widehat{p} - p}{\sqrt{\dfrac{p(1-p)}{n}}} \sim N(0,1)$$

Note: $\sqrt{p(1-p)/n}$ is the **standard error** of $\widehat{p}$. Since the value of p is unknown (as we are currently estimating it), we can not use the standard error in our confidence interval.

Instead, we use the **estimated standard error** of $\widehat{p}$, which is $\sqrt{\widehat{p}(1-\widehat{p})/n}$, and the pivotal quantity that we'll use is

$$Z = \frac{\widehat{p} - p}{\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}} \approx N(0,1)$$

The following is an approximate $100(1-\alpha)\%$ confidence interval for p:

$$\widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

Note: The validity of this depends on being able to approximate the binomial distribution with a normal distribution. Recall that to do so, we needed to have at least five successes and at least five failures.

Example 10: A sample of 1380 randomly selected books produced by a publishing company finds that 25 have bookbinding errors. Find a 95% confidence interval for p, the proportion of books with bookbinding errors.

If we are estimating sample size for population proportion, we'll find that for a given margin of error, d:

$$n = \frac{(z_{\alpha/2})^2 \widehat{p}(1-\widehat{p})}{d^2}$$

Note: To use this to calculate sample size, we need to know the value of $\widehat{p}$ before we carry out the study (which is of course impossible). In practice, we use a value from a prior study as an estimate for what $\widehat{p}$ will be.

If there has been no prior study, or if the prior estimate for p seems likely be different than the current value, there is another option:

We can show that $\widehat{p}(1-\widehat{p})$ has a maximum value of $1/4$, when $\widehat{p} = 1/2$. So the **maximum** sample size needed to guarantee a CI for p with a given margin of error is found by using:

$$n = \frac{(z_{\alpha/2})^2}{4d^2}$$

Example 11: In an earlier study, it was found that 1.4% of all microchips made by a particular manufacturer were defective. Using this as a pilot study, estimate the sample size needed to create a 99% confidence interval for p, the true proportion of defective microchips, with a margin of error of 0.005.

Example 12: We wish to carry out a telephone survey to estimate p, the proportion of island residents who want a bridge to the mainland. How many people must we call in order to estimate p with 98% confidence, to within 0.01?

Example 13: How many would we need to call in order to estimate p with 95% confidence, to within 0.025?

Example 14: The maximum safe level of bacteria per $1\ mL$ sample of water is 50. Suppose that 60 samples of water are taken, and the average bacteria content is 51.7 bacteria per sample, with a standard deviation of 7 bacteria per sample. Is there evidence that the water supply is not safe to drink?

With this question we are trying to decide between two hypotheses. One hypothesis is that the water is safe to drink. A second hypothesis is that the water is unsafe to drink. When we are trying to decide between two hypotheses, we are performing a **hypothesis test**.

In a hypothesis test, one hypothesis is called the **null hypothesis**, and is denoted H_0. We assume that the null hypothesis is true, and only reject it if there is significant evidence to the contrary.

The second hypothesis is called the **alternative hypothesis** or **research hypothesis**, and is denoted H_a or H_1.

We will first focus on the p-value approach to hypothesis testing. The **p-value** is the probability of seeing results as extreme as the observations actually made *if the null hypothesis were true*.

If the p-value is large, our observations are consistent with the null hypothesis.

If the p-value is small, then our observations are **not** consistent with the null hypothesis.

The p-value is calculated using some appropriate **test statistic**.

The p-value approach

1. Define the parameters to be tested.
2. Define H_0 and H_a.
3. Specify the test statistic and its distribution under H_0.
4. Calculate the observed value of the test statistic.
5. Calculate the p-value.
6. Report the strength of evidence against H_0:
   - *very strong* if p-value ≤ 0.01
   - *strong* if $0.01 < p$-value ≤ 0.05
   - *moderate* if $0.05 < p$-value ≤ 0.1
   - *little or none* if $0.1 < p$-value
7. Answer any other questions given (i.e. report the value of the estimate, report the value of the estimated standard error, etc.)

Example 15: The maximum safe level of bacteria per $1\ mL$ sample of water is 50. Suppose that 60 samples of water are taken, and the average bacteria content is 51.7 bacteria per sample, with a standard deviation of 7 bacteria per sample. Is there evidence that the water supply is not safe to drink?

The test we just performed is called a **one-tailed test**, as H_a consists of μ being in one region. If our H_a consists of the parameter being in one of two regions, then we call the test a **two-tailed test**.

Note: When our test statistic has a symmetric distribution (Z or t), then the p-value for a two-tailed test is exactly twice that of the equivalent one-tailed test.

Example 16: A certain medication is supposed to contain $350\ mg$ of the active ingredient per pill. It is known from previous work that this content is normally distributed with a standard deviation of $3.5\ mg$. Suppose a random sample of 5 pills are taken, and the average content is $346.4\ mg$. Is the mean pill content not $350\ mg$?

The significance-level approach

Sometimes we are asked to test H_0 at some significance level α. We carry out the hypothesis test in much the same way: defining parameters, calculating the value of the test statistic, finding the p-value.

Rather than giving the level of strength against H_0, as in the p-value approach, we instead either reject or don't reject H_0 by the following rule:

- If p-value $\leq \alpha$, then reject the null hypothesis, at the level of α.
- If p-value $> \alpha$, then do not reject the null hypothesis, at the level of α. (Some will phrase this as "fail to reject the null hypothesis")

Example 17: For the pill example, if we were asked to test our hypotheses at the level of $\alpha = 0.01$, what would our conclusion be?

Example 18: What if we were testing at the level of $\alpha = 0.05$?

The confidence interval approach (two-tailed test)

Suppose we already constructed a $(1-\alpha)100\%$ confidence interval for μ.

If we were to test $H_0 : \mu = k, \quad H_1 : \mu \neq k$, we would make our decision based on the following rule:

- If k is inside the $(1-\alpha)100\%$ confidence interval, we fail to reject the null hypothesis at the level of α.
- If k is not inside the $(1-\alpha)100\%$ confidence interval, we reject the null hypothesis at the level of α.

Example 19: Using our pill data, we can find that a 95% confidence interval for μ is $(343.77, 349.03)$.

What would our conclusion be if we test $H_0 : \mu = 344, H_1 : \mu \neq 344$ at the level of $\alpha = 0.05$?

Example 20: What would our conclusion be if we test $H_0 : \mu = 342$, $H_1 : \mu \neq 342$ at the level of $\alpha = 0.05$?

Example 21: The lengths of mourning doves (from beak to tail) are known to be normally distributed. Suppose that 5 mourning doves are selected at random, and it is found that the average length of the mourning dove is $32.4\ cm$, with a standard deviation of $2.9\ cm$.

Let μ denote the true mean length of mourning doves. Test the hypotheses $H_0 : \mu = 30$, $H_a : \mu > 30$ at the level of $\alpha = 0.1$.

Example 22: What would the conclusion be if we were testing at the level of $\alpha = 0.05$?

Example 23: In a sample of 46 people, we find the average blood glucose level upon waking up is $5.3\ mmol/L$ with a standard deviation of $1.2\ mmol/L$. Is there reason to believe that the true mean blood glucose level upon waking for people is not $5\ mmol/L$?

Example 24: Brand X cola decides that if the proportion of customers who prefer their brand ever falls below 40%, they will launch a new ad campaign. They take a survey of 150 randomly selected customers, and find that 57 prefer Brand X over the competitors.

Is there evidence that the true proportion of customers referring Brand X is less than 0.4? Test at the level of $\alpha = 0.2$

Errors and Hypothesis Tests: Two types of errors are possible in a hypothesis test. A **Type I** error (Rejection Error) is made when we reject the null hypothesis when it is true. A **Type II** error (Acceptance Error) is made when we do not reject the null hypothesis when it is false.

For the pill example, a Type I error would be made if we said that the pills didn't contain $350\ mg$ when they actually did.

A Type II error would be made if we said the pills contained $350\ mg$ when they actually didn't.

We use α and β to represent the probability of committing a Type I and Type II error, respectively.

Relationship between the errors: If we require extremely strong evidence against H_0 before we'll believe H_a, then we are attempting to make α small; as a consequence, β becomes larger.

If we are willing to accept not particularly strong evidence against H_0 before we'll believe H_a, then we are letting α be larger; as a consequence, β becomes smaller.